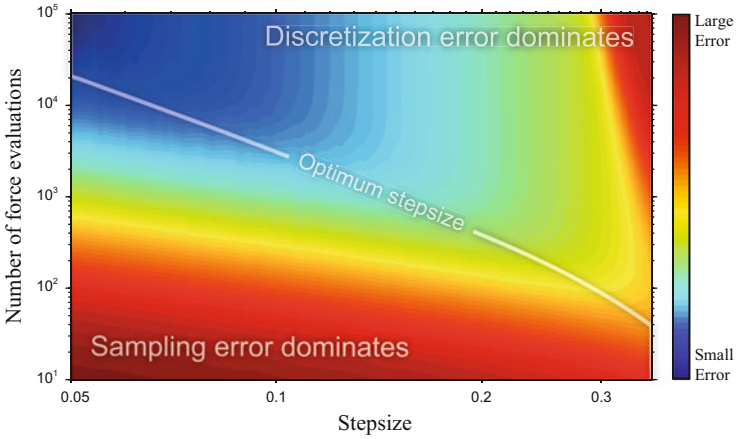


Fig. 7.2 We plot the error in the distribution of $q \in \mathbb{R}$ for $U(q) = (q^2 - 1)^2$. The error in the computed distribution is a function of both the number of force evaluations and the stepsize used, with the optimal choice of stepsize for a given number of force evaluations shown by the *white guideline*. Due to discretization bias, choosing the maximum stepsize is not always the best strategy

figure: while increasing the stepsize means we more rapidly converge with each step, the distribution that the discretization scheme converges to is incorrect. For small stepsizes, this error due to discretization may be negligible when compared to the sampling error, but we converge more slowly towards the final distribution. As the stepsize is increased, the discretization error may dominate the overall error in calculation.

7.3 Sampling Iterations

By way of illustration of what we mean by a numerical sampling method, consider the system of independent Ornstein-Uhlenbeck (OU) processes

$$\mathrm{d}\boldsymbol{p} = -\gamma \boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}. \tag{7.2}$$

Following the argument of Sect. 6.3.2 in the previous chapter,

$$\begin{aligned}\mathrm{d}[\exp(\gamma t)\boldsymbol{p}] &= \gamma \exp(\gamma t)\boldsymbol{p}\mathrm{d}t + \exp(\gamma t)[-\gamma \boldsymbol{p}\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}] \\ &= \exp(\gamma t)\sqrt{2\mathrm{k_B T}\gamma}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}.\end{aligned}$$